Pointed connected coverings are classified by subgroups of the fundamental group in homotopy type theory

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Editorial note. Copyright Samuel Mimram and Émile Oleon; official CSL2026.21, CC BY4.0. One pointed-connected-covering/subgroup classification theorem; its support/inverse directions and existing generated/presented-group delooping background do not add counts.

Allowed standard prerequisites (editorial).

- Standard HoTT truncation, long exact homotopy sequence, Grothendieck action/fibration duality and delooping background. Original paper bibliography including HoTT book and generated/presented-group delooping sources; existing lane LL022/LL023 are background only. Full current Galois-fibration and new classification/fiber proofs are actually retained.

2 Homotopy type theory

We begin by briefly recalling the main notations and tools of homotopy type theory. Detailed presentations can be found in [14, 12].

Universe.

We write $\mathcal{U}$ for the universe type, whose elements are small types, which is supposed to be closed under dependent sum and product types. Given a type $A : \mathcal{U}$ and a type family $B : A \to \mathcal{U}$, we write $\Sigma A.B$ or $\Sigma(x : A).B\,x$ for dependent sum types and $\Pi A.B$ or $\Pi(x : A).B\,x$ or $(x : A) \to B\,x$ for dependent product types. As customary, we respectively write $A \times B$ and $A \to B$ for product and arrow types, which correspond to non-dependent particular cases of the previous constructions. We respectively write 0 and 1 for the initial and terminal types.

Identities.

The type theory features a notion of definitional equality and we write $t \equiv u$ when two terms t and u are definitionally equal. It also features a notion of propositional equality: given $x, y : A$, we write $x = y$ for the type of identities or paths between x and y in A. For any point $x : A$, there is a path $\mathrm{refl} : x = x$ witnessing reflexivity. Given paths $p : x = y$ and $q : y = z$, we can construct paths $p \cdot q : x = z$ corresponding to concatenation or transitivity, and $p^{-1} : y = x$ corresponding to inverse or symmetry.

Pointed types.

A pointed type is a type A together with a distinguished element, often written $\star_A$ or even $\star$ (equivalently, the distinguished element can be specified by providing a map $1 \to A$). A pointed map $f : A \to B$ between pointed types A and B is a map between the underlying types together with an identification $f(\star_A) = \star_B$. We write $A \to_\star B$ for the corresponding type of pointed maps.

Univalence.

A map $f : A \to B$ is an equivalence when it admits both a left and a right inverse. We write $A \simeq B$ for the type of equivalences between A and B. Any identity between two types canonically induces an equivalence between them. The univalence axiom states that the corresponding map $(A = B) \to (A \simeq B)$ is itself an equivalence: in particular, any equivalence induces an identity.

Homotopy levels.

A type A is contractible when it is equivalent to 1. A type A is a proposition (resp. a set, resp. a groupoid) when for any elements $x, y : A$ the type $x = y$ is contractible (resp. a proposition, resp. a set). The type of sets is denoted Set. More generally, we can define a notion of n-type for $n \in \{-2, -1\} \cup \mathbb{N}$ by stating that a (-2)-type is a contractible one, and an $(n{+}1)$-type is a type A in which $x = y$ is an

n-type for every $x, y : A$. We write $\mathrm{isType}_n(A)$ for the predicate indicating that A is an n-type, which can be shown to be a proposition. We write $\mathcal{U}_n \equiv \Sigma\mathcal{U}.\,\mathrm{isType}_n$ for the type of n-types.

Truncation.

The n-truncation $\|A\|_n$ is the universal way of turning a type A into an n-type: it comes equipped with a map $|-|_n : A \to \|A\|_n$ such that any map $f : A \to B$, whose target B is an n-type, induces a unique map $\tilde{f} : \|A\|_n \to B$ such that $\tilde{f}(|x|_n) = f(x)$ for $x : A$. A type A is n-connected when $\|A\|_n = 1$. In particular, a type is connected (resp. simply connected) when it is 0-connected (resp. 1-connected). The connected component of an element a of A is $\Sigma(x : A).\|a = x\|_{-1}$.

Loop space.

The circle S^1 is the free pointed type containing a path $\mathrm{loop} : \star = \star$. Given a pointed type A, its loop space $\Omega\, A$ is $\star = \star$, which can be shown to coincide with the type $\mathrm{S}^1 \to_\star A$. Its fundamental group is the type $\pi_1(A) \equiv \|\,\Omega\, A\|_0$, which is canonically equipped with a group structure induced by path concatenation. When A is a pointed connected groupoid, its loop space coincides with its fundamental group, so that it is a group. A delooping of a group G is a type $\mathrm{B}\, G$ equipped with an isomorphism of groups $\Omega\,\mathrm{B}\, G \cong G$ (such a type always exists and is unique, thus the notation). It is easily shown that a map $\mathrm{B}\, G \to \mathrm{Set}$ corresponds to a set equipped with an action of G in the usual sense.

Fiber sequences.

Given a map $f : A \to B$ and $y : B$, the fiber of f at y is the type $\mathrm{fib}_f\, y \equiv \Sigma(x : A).(f(x) = y)$. When B is pointed, the kernel of f is $\ker f \equiv \mathrm{fib}_f \star$. A composable pair of morphisms

$$F \longrightarrow A \xrightarrow{f} B$$

is a fiber sequence when F is the kernel of f and the map $F \to A$ is the first projection. A map is n-connected (resp. n-truncated) when all its fibers are.

The Grothendieck duality.

A fundamental property in homotopy type theory is that, given a type A, fibrations over A correspond both to types over A and to type families indexed by A: this is the Grothendieck duality. We write $\mathcal{U}/A \equiv \Sigma(B : \mathcal{U}).(B \to A)$ for the type of types over A and $A \to \mathcal{U}$ for the type of type families indexed by A. We have a function $\mathrm{fib} : \mathcal{U}/A \to (A \to \mathcal{U})$, which to $p : B \to A$ associates the fiber $\mathrm{fib}_p : A \to \mathcal{U}$, and a function $\int : (A \to \mathcal{U}) \to (\mathcal{U}/A)$, which to a family $F : A \to \mathcal{U}$ associates the first projection map $\pi : \Sigma A.F \to A$. The following is shown in [14, Section 4.8]:

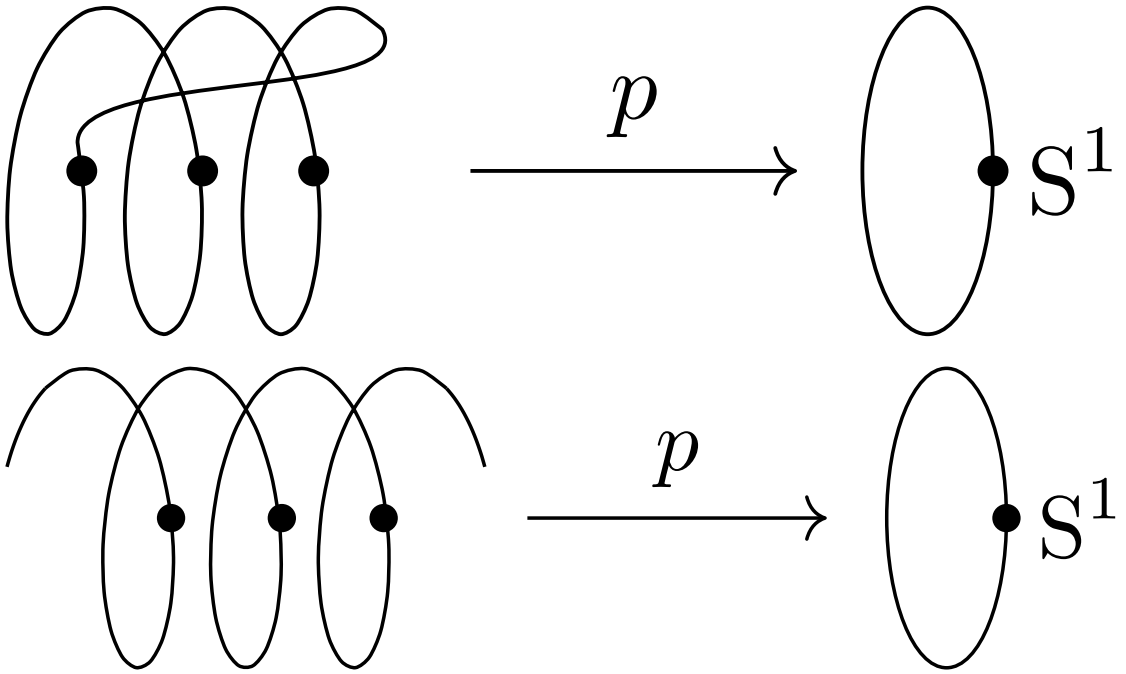
Theorem 1.

Given a type A, the above functions induce an equivalence $\mathcal{U}/A \simeq (A \to \mathcal{U})$. Moreover, this correspondence is functorial in the sense that given $p : B \to A$ and $q : C \to A$, a morphism $f : B \to C$ with $q \circ f = p$ corresponds to a family of maps $(x : A) \to \mathrm{fib}_p\, x \to \mathrm{fib}_q\, x$ in a way which preserves identities and composition (and thus equivalences).

3 Higher covering types

3.1 Covering spaces in topology

We briefly recall here the traditional notion of covering space in topology and refer to standard textbooks for details [7]. Given a topological space A, a covering is a space B together with a map $p : B \to A$ which is locally trivial. This means that every point $x : B$ admits an open neighborhood such that the preimage $p^{-1}(U)$ is homeomorphic to a space of the form $U \times F$, where F is a set equipped with the discrete topology and the restriction of p to $p^{-1}(U)$ is the first projection. When B is connected, which will be the case of interest here, the cardinal of F has to be the same for every point x and is called the number of sheets of the covering. Below we have figured a covering with 3 sheets (on the left) and with countably many sheets (on the right):



In order for the constructions of coverings to be well-behaved, we will implicitly assume in the following that A satisfies reasonable assumptions (namely being connected, locally path-connected, semilocally simply-connected). We will also assume that it comes equipped with a distinguished point $\star$. An important feature of coverings is that they have the path lifting property: given a path $\pi : x \rightsquigarrow y$ in A and an element $\tilde{x} \in \tilde{A}$ with $p(\tilde{x}) = x$ there is a unique path $\tilde{\pi} : \tilde{x} \to \tilde{y}$ with $p(\tilde{\pi}) = \pi$. This implies that we have an action of the fundamental group $\pi_1(A)$ on the fiber $p^{-1}(\star)$ and, in fact, this characterizes coverings:

Proposition 2 ([7, Section 1.3, p. 70]).
Coverings of A are in bijection with sets equipped with an action of $\pi_1(A)$.

The universal covering is the only covering $p : \tilde{A} \to A$ with $\tilde{A}$ simply connected. For instance, the universal covering of S^1 is the "helix" pictured on the right above. This space can be shown to be unique up to isomorphism and can be constructed as follows:

Proposition 3 ([7, Section 1.3, p. 64]).
The universal cover $\tilde{A}$ of A can be constructed as the space whose points are pairs consisting of a point $x : A$ and a path $p : \star \rightsquigarrow x$ up to homotopy, equipped with a suitable topology, the map $p : \tilde{A} \to A$ being given by first projection.

The fundamental group $\pi_1(A)$ of A is the group whose elements are homotopy classes of paths $\star \rightsquigarrow \star$ in A, with concatenation as multiplication and constant paths as neutral elements. By the construction of Proposition 3, given $x : A$, the fiber $p^{-1}(x)$ is $\pi_1(A)$, and we have an action of $\pi_1(A)$ on this fiber given by left multiplication, which induces an action of $\pi_1(A)$ on $\tilde{A}$.

Proposition 4 ([7, Theorem 1.38]).
The quotient $\tilde{A}/\pi_1(A)$ of the universal cover under the above action is precisely A.

The action can be shown to be free, so that the quotient above coincides with the homotopy quotient.

3.2 The universal fibration

We begin by describing a situation in homotopy type theory, which is close to coverings, and fundamental with respect to the characterization of identity types. This construction might seem a bit artificial at first, but we will see that the point of view nicely generalizes to coverings, in the sense that what we describe in this section is a kind of "universal ∞-covering".

Suppose given a pointed type A. A pointed type over A is a pointed morphism $p : B \to_\star A$. A morphism between two such pointed types $p : B \to_\star A$ and $q : C \to_\star A$ is a map $f : B \to C$ together with an equality $f \circ p = q$. Such morphisms compose in the expected way, which is compatible with the composition of underlying maps.

Definition 5.
A pointed type $p : B \to_\star A$ is universal when for every pointed type $q : C \to A$, the type of morphisms from p to q is contractible.

Since it satisfies a universal property, the universal pointed type is unique, and it exists thanks to the following characterization:

Proposition 6.
A pointed type $p : B \to_\star A$ is universal if and only if B is contractible.

Proof.
By definition of morphisms and initiality of 1, the type 1 is initial among pointed types over A. A universal pointed type $p : B \to A$ is also initial among pointed types over A, by the universal property, and thus B is equivalent to 1, i.e. contractible. ◀ By immediate computations, we have:

Proposition 7.
The fiber of the universal pointed type $p : 1 \to A$ is $\Omega\, A$.

It is very illuminative to translate the previous definitions and properties under the Grothendieck duality (see Theorem 1). A pointed type over A corresponds to a fibration $P : A \to \mathcal{U}$ together with a distinguished element $\star_P$ of $P\star$, and, by Proposition 6, such a fibration is universal precisely when the total space $\Sigma A.P$ is contractible. The universal pointed type thus corresponds to the universal fibration, which is the map $\mathcal{F}_A : A \to \mathcal{U}$ defined by $\mathcal{F}_A\, x \equiv (\star = x)$, and pointed by $\mathrm{refl}_\star$. By Theorem 1, a morphism between fibrations $P, Q : A \to \mathcal{U}$ corresponds to a family of maps $f : (x : A) \to P\, x \to Q\, x$ together with an identification $f \star \star_P = \star_Q$. The initiality property of universal pointed types (Definition 5) then translates as the fundamental theorem of identity types [12, Theorem 11.2.2]:

Theorem 8.
Suppose given a type family $P : A \to \mathcal{U}$ pointed by $\star_P : P\star$, together with a family of maps

$$F : (x : A) \to (\star = x) \to P\, x$$

and an identification $F \star \mathrm{refl}_\star = \star_P$. Then F is a family of equivalences if and only if the total space $\Sigma A.P$ is contractible.

Proof.
Under the Grothendieck duality of Theorem 1, the type family P corresponds to the pointed type $p : \Sigma A.P \to A$ over A, given by the first projection. The family of maps F then corresponds to a morphism between the universal type over A and p, i.e. to a pointed map $f : 1 \to \Sigma A.P$. Then F is a family of equivalences if and only if the induced map f is an equivalence, i.e. if and only if $\Sigma A.P$ is contractible. ◀

▶ **Remark 9.**
In the situation of the above theorem, the map P can be thought of as being representable, in the sense that $Px = (\star = x)$ (and identities can be thought of as homs in types). This notion was actually used by Voevodsky [15] in order to first define identity types.

3.3 Higher covering types

We now introduce the notion of n-covering type, for any natural number n, which can be understood as an n-truncated variant of the pointed types of previous section. The traditional notion of covering is the particular case $n = 0$, as we indicate in remarks.

An n-covering of A is a map $p : B \to A$ whose fibers are n-types; such a map is also said to be n-truncated. We write

$$\mathrm{Covering}_n(A) \quad \equiv \quad \Sigma(B : \mathcal{U}).\Sigma(f : B \to A).(x : A).\, \mathrm{isType}_n(\mathrm{fib}_f\, x)$$

for the type of coverings of A. Under the Grothendieck duality, those can also be defined as families of n-types.

Lemma 10.
We have an equivalence $\mathrm{Covering}_n(A) \simeq (A \to \mathcal{U}_n)$.

Proof.
Follows immediately from Grothendieck duality (Theorem 1). ◀

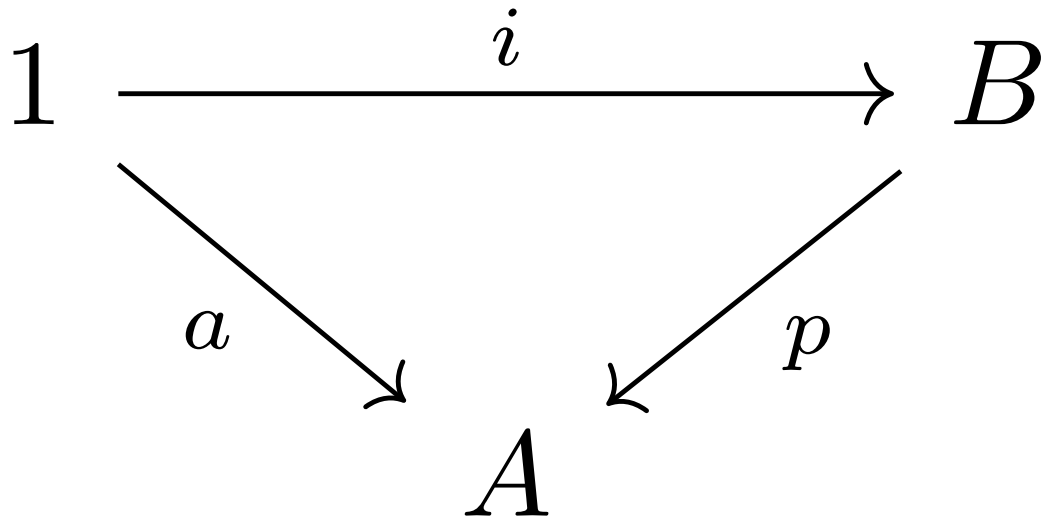
▶ **Remark 11.**
For $n = 0$, we recover the definition covering types of [8, Definition 1], as maps $A \to \mathrm{Set}$. Since Set is a groupoid [14, Theorem 7.1.11], the universal property of groupoid truncation provides us with

an equivalence $(A \to \mathrm{Set}) \simeq (\|A\|_1 \to \mathrm{Set})$. A covering of A thus corresponds to a map $\|A\|_1 \to \mathrm{Set}$. Moreover, we have by Proposition 30 that $\|A\|_1$ is a $\mathrm{B}\,\pi_1 A$, from which deduce that a covering of A corresponds to a set equipped with an action of $\pi_1 A$. We thus recover the traditional Proposition 2, which was formalized in homotopy type theory in [8, Theorem 4].

A morphism f between n-coverings $p : B \to A$ and $q : C \to A$ is a map $f : B \to C$ together with an equality $p = q \circ f$.

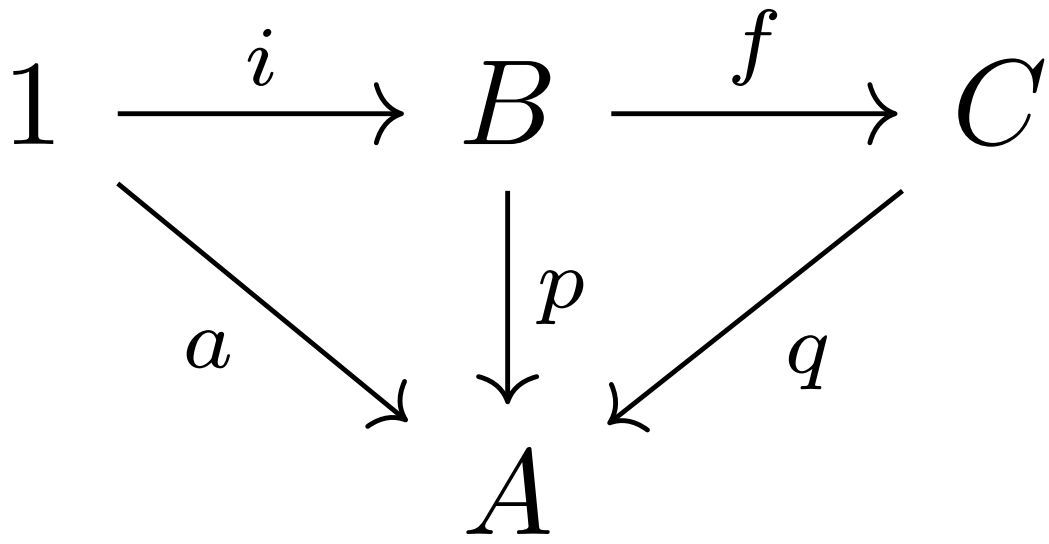
A pointed n-covering is a pointed map $p : B \to_\star A$ between pointed types, whose underlying map is an n-covering. This corresponds exactly to the following notion:

Definition 12.
A pointed n-covering is a factorization



of the pointing map $a : 1 \to A$ as $a = p \circ i$ where p is n-truncated.

In the following, we sometimes assimilate the covering to the map $p : B \to A$ leaving the data of the pointing $i : 1 \to B$ of B implicit. As expected, a morphism f between pointed n-coverings p and q is a pointed map which is a morphism between the underlying n-coverings, i.e. such that the factorization of the target is $q \circ (f \circ i)$:



Definition 13.

A pointed n-covering $p : B \to A$ is universal when for every n-covering $q : C \to A$ there exists a unique map $f : B \to C$ of pointed n-coverings.

▶ **Remark 14.**

For $n = 0$, we recover the characterization of pointed universal coverings as being initial in the category of pointed coverings [8, Lemma 12].

We sometimes write $\tilde{A}$ for the universal n-covering. This is justified by the fact that, being defined by a universal property, it is uniquely characterized:

Lemma 15.

Any two universal pointed n-coverings are uniquely isomorphic.

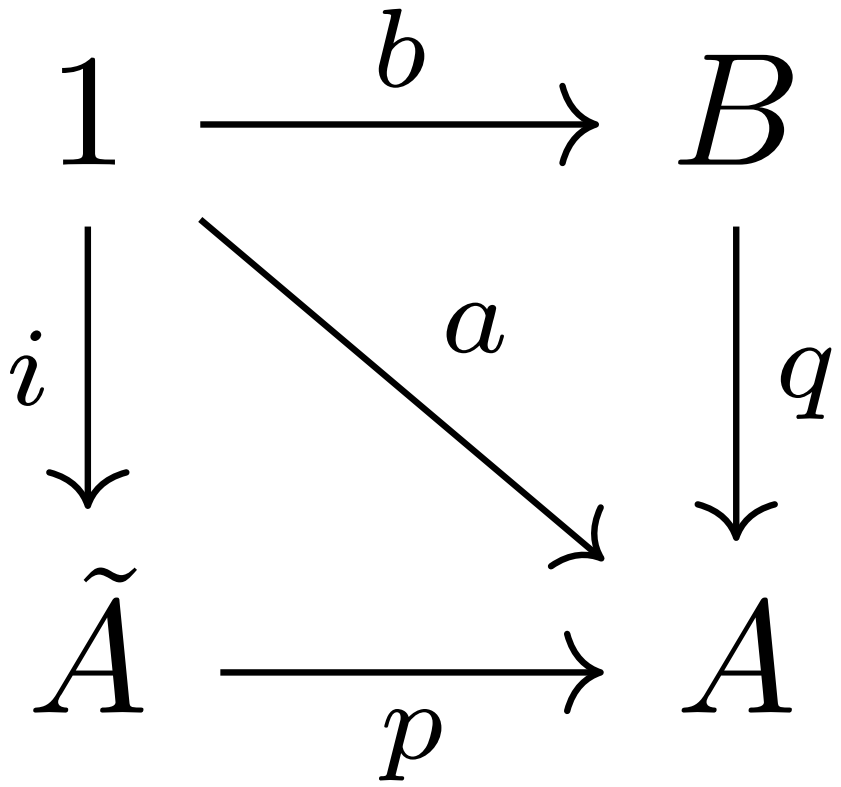
We have the following characterization of universal n-coverings:

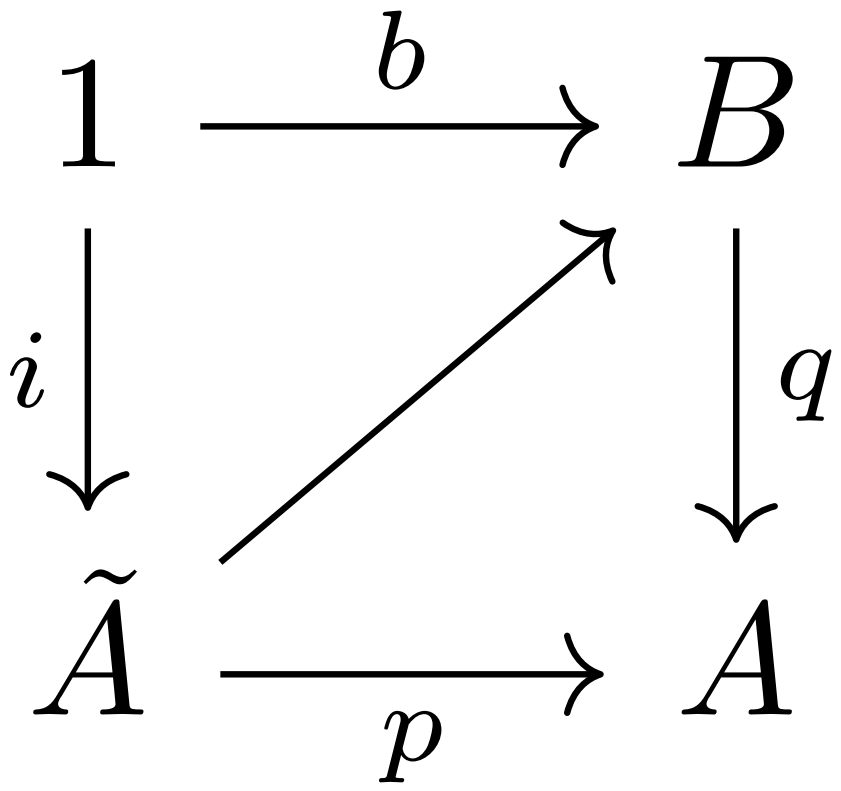
Theorem 16.

Given a pointed type A with pointing map $a : 1 \to A$, any factorization $a = p \circ i$ as n-connected map i and followed by an n-truncated map p exhibits p as a universal n-covering. Moreover, such a factorization always exists and is unique.

Proof.

By [14, Theorem 7.6.6], the map $a : 1 \to A$ admits a unique factorization $a = p \circ i$ as required. Moreover, given a n-covering $q : B \to A$ pointed by $b : 1 \to B$, we have a commuting square as on the left





The commutation of the two triangles being given by the fact that we have two pointed n-coverings of A pointed by a. By [14, Theorem 7.6.7], because i is n-connected and p is n-truncated, there is a unique map $\tilde{A} \to B$ making the two triangles on the right commute, and p is thus universal in the sense of Definition 13. ◀

In practice, the universal n-covering can be constructed as follows. We recall from [14, Definition 7.6.3] that, given a map $f : B \to A$ and a natural number n, its n-image is

$$\mathrm{im}_n\, f \quad \equiv \quad \Sigma(x : A).\| \mathrm{fib}_f\, x\|_n$$

We write $i_n : B \to \mathrm{im}_n\, f$ for the canonical map such that $i_n(x) = (f(x), |\,\mathrm{refl}_x\,|_n)$ and $p_n : \mathrm{im}_n\, f \to A$ for the first projection map.

Proposition 17.
The factorization $a = p_n \circ i_n$ of the pointing map a as above, exhibits $p_n : \mathrm{im}_n\, a \to A$ as the universal covering of A.

Proof.
By [14, Lemma 7.6.4], this factorization satisfies the conditions of Theorem 16. ◀

► **Remark 18.**
For $n = 0$, we recover the usual definition of the universal covering space as the type of homotopy classes of paths from the distinguished point:

$$\tilde{A} \quad = \quad \Sigma(x : A).\|\star = x\|_0$$

For $n = -1$, the universal covering of A is its set of connected components. Finally, for $n = \infty$ (we adopt the convention that $\|A\|_\infty \equiv A$), we recover the universal pointed type of Section 3.2 by contractibility of singletons [14, Lemma 3.11.8].

Example 19.

The universal covering of $A \equiv \mathrm{S}^1$ is $\Sigma(x : A).(\star = x)$ (we can remove the 0-truncation because S^1 is a groupoid) and thus contractible by [14, Lemma 3.11.8] (see Lemma 26 for a generalization of this argument).

▶ **Remark 20.**
The factorization results used above hold more generally for any map $a : B \to A$ where B is not necessarily contractible. In this sense, given an arbitrary map $a : B \to A$, we can think of $\mathrm{im}_n\, a$ as the "universal n-cover of A relative to a".

The universal n-covering can also be characterized as the covering whose total space is $(n{+}1)$-connected. This can be shown using the following lemma proved in appendix.

Lemma 21.
A pointed connected type A is $(n{+}1)$-connected (resp. $(n{+}1)$-truncated) if and only if the pointing map $1 \to A$ is n-connected (n-truncated).

Theorem 22.
Given a pointed connected type A, the universal pointed n-covering is the $(n{+}1)$-connected pointed n-covering of A.

Proof.
By Definition 12, a pointed n-covering $p : B \to A$ is a factorization of the pointing map $a : 1 \to A$ as $a = p \circ i$ with p n-truncated. By Theorem 16, it is universal if and only if $i : 1 \to B$ is n-connected which, by Lemma 21 is equivalent to the fact that B is $(n{+}1)$-connected. ◀

▶ **Remark 23.**
For $n = 0$, we recover the fact that the universal covering is the only 1-connected covering of a type [8, Lemma 11].

We now formalize the intuition that the universal n-covering $\tilde{A}$ of A provides a way to "kill" all the homotopy in dimension $i \leq n + 1$. This is based on what we call the fundamental fibration associated to the universal n-covering:

Theorem 24.
Writing $\tilde{A}$ for the universal n-covering, we have a fiber sequence

$$\tilde{A} \xrightarrow{\quad p \quad} A \xrightarrow{\quad |-|_{n+1} \quad} \left\| A \right\|_{n+1}$$

Proof.
We have

$$\ker |-|_{n+1} \quad \equiv \quad \Sigma(x : A).(|\star|_{n+1} = |x|_{n+1}) \quad = \quad \Sigma(x : A).\|\star = x\|_n \quad = \quad \tilde{A}$$

where middle equality is [14, Theorem 7.3.12] and right one is Proposition 17. ◀ This was actually taken to be the definition of n-coverings in [2]. As a corollary, we have the following characterization of the homotopy groups of the universal n-covering:

Proposition 25.
We have $\pi_i(\tilde{A}) = 1$ for $i \leq n + 1$ and $\pi_i(\tilde{A}) = \pi_i(A)$ for $i > n + 1$.

In particular, this suggests that the universal covering should be contractible when A has no homotopy in dimension $i > n + 1$:

Lemma 26.
Given a $(n{+}1)$-truncated pointed type A, its universal n-covering is contractible.

Proof.
Since A is supposed to be $(n{+}1)$-truncated, the pointing map $a : 1 \to A$ is n-truncated by Lemma 21 and the factorization $a = a \circ \mathrm{id}_1$ of the pointing map as an n-connected map followed by an n-truncated map has to be the factorization of the universal covering (Theorem 16) by uniqueness. ◀

We would like to end this section with the following conjecture, which would allow constructing deloopings of higher groups based on the previous construction. Note that, below, the coverings are not supposed to be pointed.

Conjecture 27.
Given a connected type A, the connected component of $\tilde{A}$ in $\mathrm{Covering}_n(A)$ is $\|A\|_{n+1}$.

Provided that this conjecture holds, the connected component of $\tilde{A}$ in A would be a pointed $(n+1)$-connected groupoid, which can thus be thought of as a delooping of the fundamental n-group of A. The following proposition shows that we have right underlying type (it would remain to be shown that we have the right higher operations for the n-group):

Proposition 28.
Given a pointed connected type A, the type of automorphisms of the universal n-covering $\tilde{A}$ is $\|\,\Omega\,A\|_n$.

Proof.
By universal property of $\tilde{A}$ (Definition 5), an automorphism $f : \tilde{A} \to \tilde{A}$ is uniquely determined by $f\star$ which is an element of $\mathrm{fib}_p\,\star$. The type of automorphisms of $\tilde{A}$ is thus $\mathrm{fib}_p\,\star$. Now, by Proposition 17, we can consider that p is the first projection $p : \Sigma(x : A).\|\,\mathrm{fib}_a\,x\|_n \to A$ whose fiber at $\star$ is $\|\,\mathrm{fib}_a\,\star\|_n$ by [14, Lemma 4.8.1], i.e. $\|\,\Omega\,A\|_n$. ◀

▶ **Remark 29.**
Let us explain why the conjecture does hold in the case $n = 0$. Writing

$$\mathrm{Comp}(p_0) \quad \equiv \quad \Sigma(p : \mathrm{Covering}(A)).\|p_0 = p\|_{-1}$$

for the connected component of the universal 0-covering $p_0 : \tilde{A} \to A$, we have

$$\Omega_{p_0}\,\mathrm{Comp}(p_0) = \Omega_{p_0}\,\mathrm{Covering}(A) = \|\,\Omega\,A\|_0 \equiv \pi_1(A)$$

where the first equality follows from the fact that the canonical projection map from $\mathrm{Comp}(p_0)$ to $\mathrm{Covering}(A)$ is an embedding [14, Lemma 7.6.4], and the second one is due to Proposition 28. Moreover, this identity is compatible with the group structures on both sets.

Given a group G, consider a delooping $A \equiv \mathrm{B}\,G$. By Grothendieck duality, the type of coverings of $\mathrm{B}\,G$ coincide with maps $\mathrm{B}\,G \to \mathrm{Set}$ (see Remark 11), i.e. with the type Set_G of sets equipped with an action of G [8, Theorem 4]. Moreover, under this identification, the universal covering corresponds to the principal G-set P_G, which is the set G equipped with the canonical action induced by right multiplication. Indeed, $\mathrm{B}\,G$ being a 1-truncated pointed type, its universal covering is contractible by Lemma 26 and is thus the pointing map $p : 1 \to \mathrm{B}\,G$, and the corresponding map $\phi : \mathrm{B}\,G \to \mathrm{Set}$ is $\phi \equiv \mathrm{fib}_p \equiv (x \mapsto \star = x)$. In particular, we have $\phi(\star) = (\star = \star) = \Omega\,\mathrm{B}\,G = G$. Moreover, given an element a of G, seen as path $a : \star = \star$, we have $(\phi^{=}a)^{\rightarrow}(b) = ab$ by the formula for transport in identity types [14, Lemma 2.11.2]. We thus have that the connected component of the principal G-set

$$\mathrm{Comp}(P_G) \equiv \Sigma(X : \mathrm{Set}_G).\|P_G = X\|_{-1}$$

is a delooping of G. This type is known as the type of G-torsors [1, 3, 6, 16].

4 The Galois correspondence

4.1 The Galois fibration

From now on, we restrict ourselves to the case $n = 0$ of n-coverings. In order to define the Galois correspondence, we first need to define the action of the fundamental group $\pi_1(A)$ of a pointed connected type A on its universal cover $\tilde{A}$. This means that we want to define a map $F : \mathrm{B}\,\pi_1 A \to \mathcal{U}$ such that $F\star = \tilde{A}$. By the Grothendieck duality (Theorem 1), this amounts to define a map $f : X \to \mathrm{B}\,G$, for some type X, whose fiber is $\tilde{A}$. Moreover, the source X has to be the homotopy quotient of $\tilde{A}$ under this action, which is known to coincide with the strict quotient because the action is free, and should thus be A itself by Proposition 4, see Section 3.1. Another important observation, is that we have a very convenient model for $\mathrm{B}\,\pi_1 A$, namely:

Proposition 30.
Given a pointed connected type A, we have that $\|A\|_1$ is a $\mathrm{B}\,\pi_1 A$.

Proof.

We take $|\star|_1$ to be the distinguished point of $\|A\|_1$. Connectedness is preserved by truncation: we have $\|\|A\|_1\|_0 = \|A\|_0 = 1$ (the first equality is [14, Lemma 7.3.15] and the second one is the fact that A is connected) and thus $\|A\|_1$ is connected. Finally, we have $\Omega\,\|A\|_1 = \|\,\Omega\,A\|_0 \equiv \pi_1 A$ by [14, Corollary 7.3.13]. ◀ The previous discussion suggests defining:

Definition 31.

Given a pointed connected type A, the associated Galois fibration is the map $|-|_1 : A \to \|A\|_1$, which we write g_A in the following.

Its target is $\mathrm{B}\,\pi_1 A$ by Proposition 30 and we have the expected fiber as the case $n = 0$ of Theorem 24:

Proposition 32.

We have $\ker g_A = \tilde{A}$, i.e. we have a fiber sequence

$$\tilde{A} \longrightarrow A \xrightarrow{g_A} \mathrm{B}\,\pi_1 A$$

.

As an interesting immediate consequence of this result, we recover the fact that the fibers of the universal covering are the fundamental group:

Proposition 33.

We have a fiber sequence

$$\pi_1 A \longrightarrow \tilde{A} \longrightarrow A$$

.

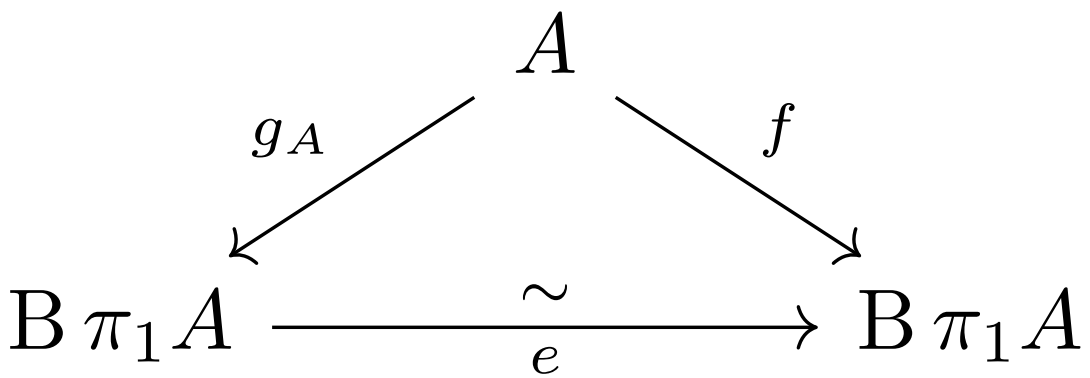
Proof.

By [14, Section 8.4], the fiber sequence of Proposition 32 can be extended on the left by $\Omega\,\mathrm{B}\,\pi_1 A$, which is $\pi_1 A$ by definition of the delooping. ◀

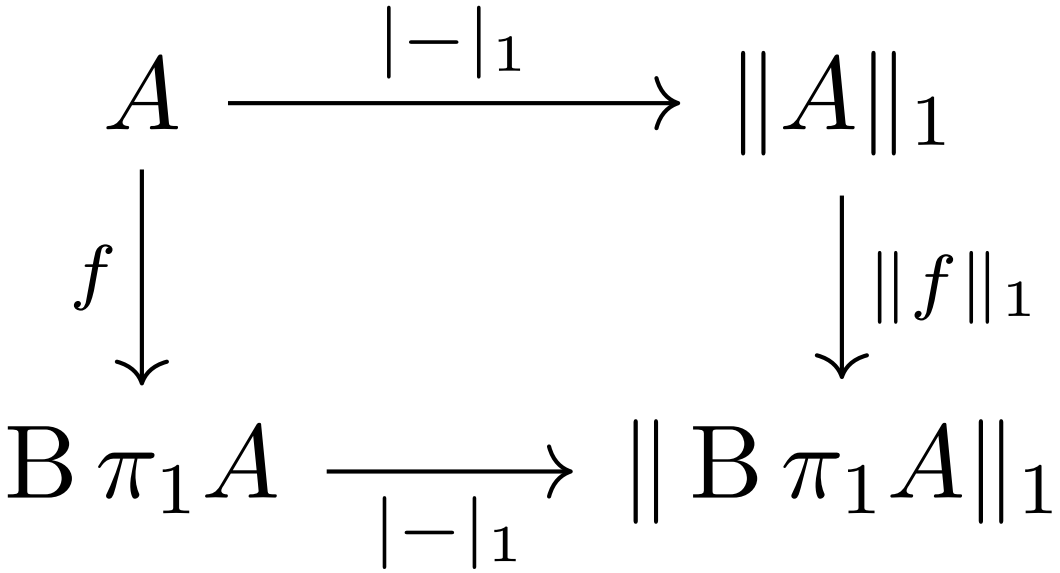
A careful reader could wonder why the action encoded by Proposition 32 is actually the "right" one in the sense that it corresponds to the traditional one in topology. We can expect that there are other such actions, i.e. maps $f : A \to \mathrm{B}\,\pi_1 A$ with $\ker f = \tilde{A}$ (what will be important here is that $\tilde{A}$ is 1-connected by Remark 23). In fact, it turns out is only one possible such action, up to an automorphism of $\mathrm{B}\,\pi_1 A$:

Proposition 34.

Given a pointed connected type A and 1-connected map $f : A \to \mathrm{B}\,\pi_1 A$, there is an equivalence $e : \mathrm{B}\,\pi_1 A \to \mathrm{B}\,\pi_1 A$ for which there is a commuting triangle



Proof.
By naturality of truncation [14, equation (7.3.4)], we have a commutative square



By definition, the upper map is g_A. The lower map is $g_{\mathrm{B}\,\pi_1 A}$, which is an equivalence because $\mathrm{B}\,\pi_1 A$ is a groupoid [14, Corollary 7.3.7]. The right vertical map is also an equivalence by [14, Lemma 7.5.14], because f is supposed to be 1-connected. Finally, $e \equiv g^{-1}_{\mathrm{B}\,\pi_1 A} \circ \|f\|_1$ is an equivalence as a composite of equivalences. ◀ The situation above is a bit subtle. It states that a 1-connected map as f has to be the Galois fibration. However, this is up to an automorphism of $\mathrm{B}\,\pi_1 A$, which might itself bear some information. Indeed, pointed automorphisms of $\mathrm{B}\,\pi_1 A$ correspond to group automorphisms of $\pi_1 A$ which might be non-trivial. In all the applications below, insights from the corresponding constructions in algebraic topology allow us to make sure that we indeed have the right action.

4.2 The Galois correspondence

In algebraic topology, we have seen in Proposition 4 that, given a nice pointed space A, there is an action of its fundamental group on its universal covering. This is in fact the basis of a classification of (connected) coverings: those correspond to the subgroups of the fundamental group, see for instance [7, Theorem 1.38]. In homotopy type theory, the action is encoded by the Galois fibration (as seen with Proposition 32) and our aim is now to develop a similar classification of coverings. In this section, we write Covering(A) for the coverings of A which are pointed and connected, i.e. whose total space is connected. Given a group G, we also write Subgroup(G) for the type of subgroups of G, i.e. injective maps $i : H \to G$ for some group H.

The following lemma, whose proof is provided in appendix, will be useful to show the classification.

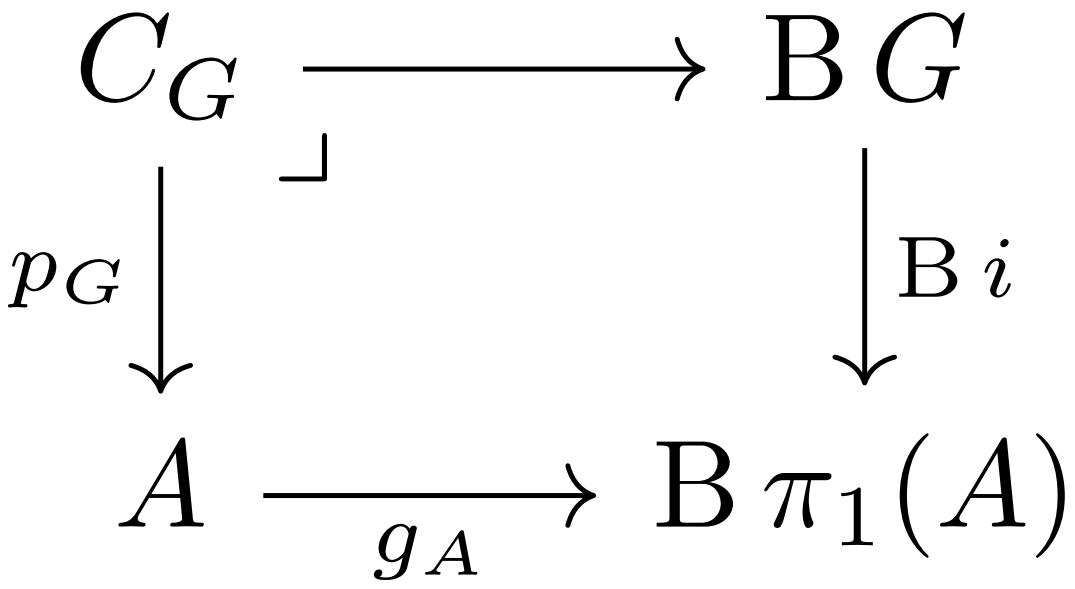
Lemma 35.
Given a subgroup $i : H \to G$, the fiber of $\mathrm{B}\,i$ merely is the set G/H.

One of the main results of this paper is the following theorem, see [13] for a formalization:

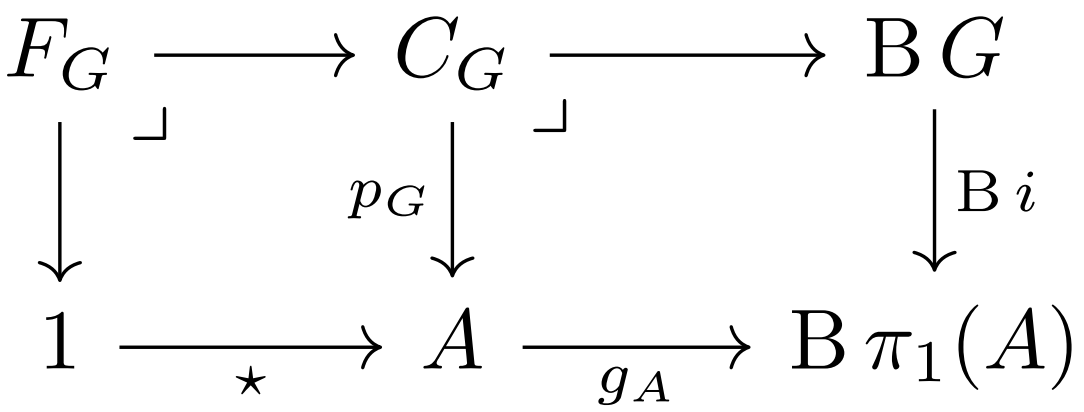
Theorem 36.
There is an equivalence between subgroups of $\pi_1(A)$ and pointed connected coverings of A:

$$\mathrm{Subgroup}(\pi_1(A)) \simeq \mathrm{Covering}(A)$$

Proof.
Given a subgroup $i : G \hookrightarrow \pi_1(A)$, the corresponding covering X is obtained as pullback of the Galois fibration along the delooping of group inclusion:



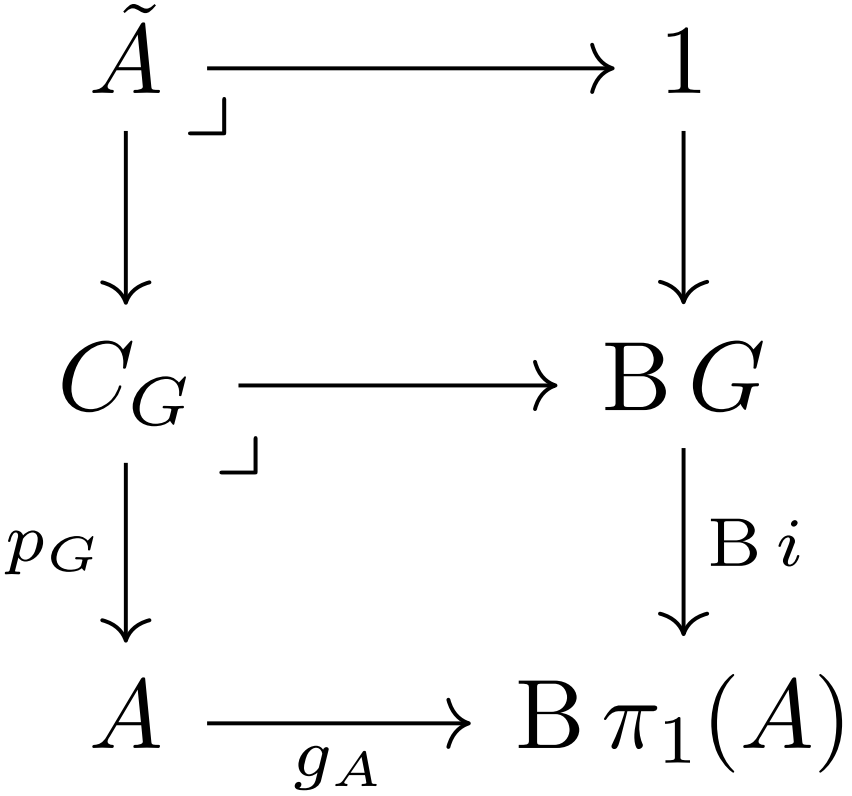
We need to show that C_G is a connected covering, i.e. it is connected and p_G has 0-truncated fibers. The fiber of the covering is given by pasting pullbacks



and thus we have

$$F_G \equiv \ker p_G = \ker \mathrm{B}\,i = \pi_1(A)/G$$

by Lemma 35 (to be precise, we only have the existence of such an equality, which is sufficient for our purposes exposed in next sentence). As a consequence, the fibers of p_G are sets. Also, by vertical pasting of pullbacks,



we have that the upper square is a pullback, i.e. a fibration sequence of the form

$$\tilde{A} \longrightarrow C_G \longrightarrow \mathrm{B}\,G$$

which describes, by action-fibration duality, an action of G on $\tilde{A}$ such that

$$C_G = \tilde{A}/\!\!/G$$

Therefore, C_G is connected as a homotopy quotient of a connected space.
Conversely, given a connected covering $f : X \to A$, we have

$$\pi_1(f) : \pi_1(X) \to \pi_1(A)$$

This is a mono, because we have the long exact sequence associated to the fibration:

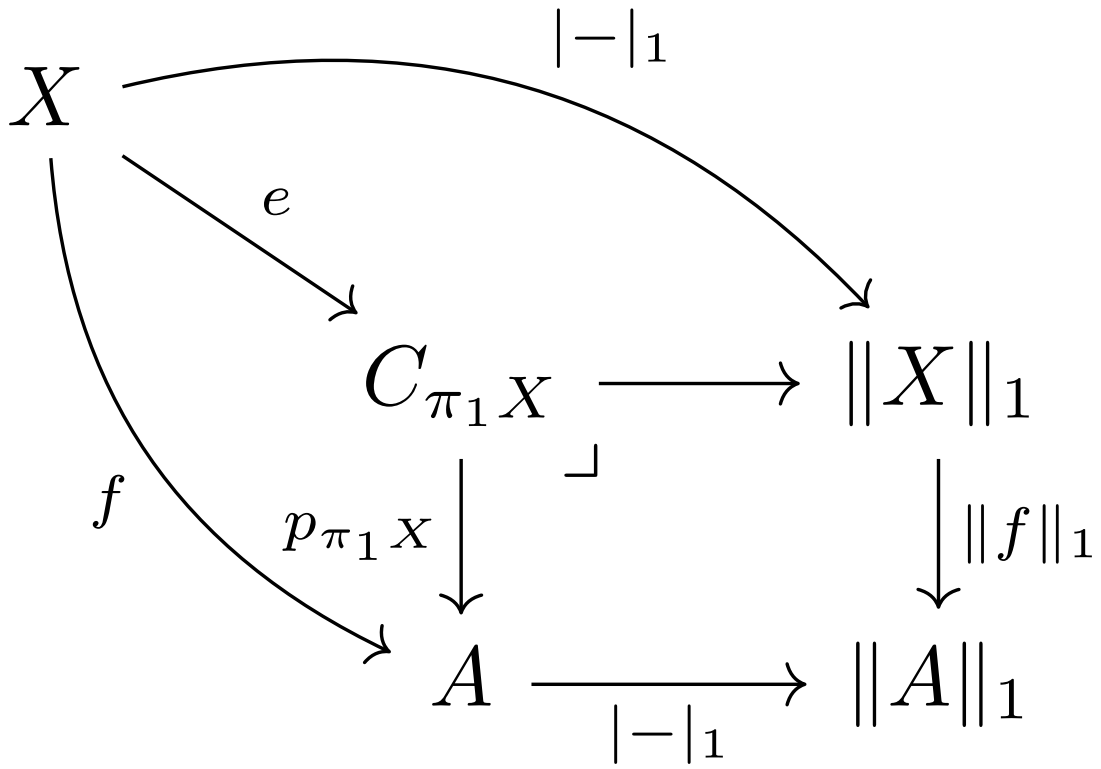
$$\pi_1(F) \longrightarrow \pi_1(X) \xrightarrow{\pi_1(f)} \pi_1(A)$$

where $F \equiv \ker f$ is a set (because f is covering) and thus $\pi_1(F) = 1$.

We now have to show that these two operations are mutually inverse. Given a connected covering $f : X \to A$, the associated connected covering (by performing the two operations) is obtained as the pullback on the left, which can be rewritten as on the right.

$$\begin{array}{ccc} C_{\pi_1 X} & \longrightarrow & \mathrm{B}\,\pi_1 X \\ {\scriptstyle p_{\pi_1 X}}\downarrow & \lrcorner & \downarrow{\scriptstyle \mathrm{B}\,\pi_1 f} \\ A & \xrightarrow[g_A]{} & \mathrm{B}\,\pi_1 A \end{array} \qquad \begin{array}{ccc} C_{\pi_1 X} & \longrightarrow & \|X\|_1 \\ {\scriptstyle p_{\pi_1 X}}\downarrow & \lrcorner & \downarrow{\scriptstyle \|f\|_1} \\ A & \xrightarrow[|-|_1]{} & \|A\|_1 \end{array}$$

The outer square of the diagram below commutes by naturality of the unit of the truncation, and we thus have a universal map $e : X \to C_{\pi_1 X}$ such that



More explicitly,

$$C_{\pi_1 X} \quad \equiv \quad \Sigma(a : A).\Sigma(y : \|X\|_1).(|a|_1 = \|f\|_1(y))$$

and $e(x) = (f(x), |x|_1, \mathrm{refl}_{|f(x)|_1})$. Because the lower-left triangle commutes, e is a map between types over A, and thus corresponds by Grothendieck duality (Theorem 1) to a family of fiberwise maps

$e_a : \mathrm{fib}_f\, a \quad \to \mathrm{fib}_{p_{\pi_1 X}}\, a$
$(x, p) \quad \mapsto (f(x), |x|_1, \mathrm{refl}, p)$

indexed by $a : A$, with $p : a = f(x)$. Let us consider the case $a \equiv \star$. We have

$$\mathrm{fib}_{p_{\pi_1 X}}\, \star \quad \equiv \quad \Sigma(a : A).\Sigma(y : \|X\|_1).\Sigma(q : |a|_1 = \|f\|_1(y)).(\star = a)$$

We can construct an inverse map to $e_\star$

$e'_\star : \mathrm{fib}_{p_{\pi_1 X}}\, \star \quad \to \mathrm{fib}_f\, \star$
$(a, y, q, p) \quad \mapsto (x, p \cdot \tilde{q})$

where we assume that $y = |x|_1$ because we are eliminating to $\mathrm{fib}_f\, \star$ which is a set (and thus a groupoid). Above, we have $p : |a|_1 = \|f\|_1(x)$, which is equivalent to $\|a = f(x)\|_0$ (by [14, Theorem 7.3.12]) and we can thus suppose that $q = |\tilde{q}|_0$ with $\tilde{q} : a = f(x)$ because we are eliminating to a set. We have

$$e' \circ e(x, p) = (x, p \cdot \mathrm{refl}) = (x, p)$$

Conversely,

$$e \circ e'(a, |x|_1, |q|_0, p) = e(x, p \cdot q) = (f(x), |x|_1, \mathrm{refl}, p \cdot q)$$

(we can suppose that the second and third arguments are truncations as above, because we are eliminating to a set). We thus have to show

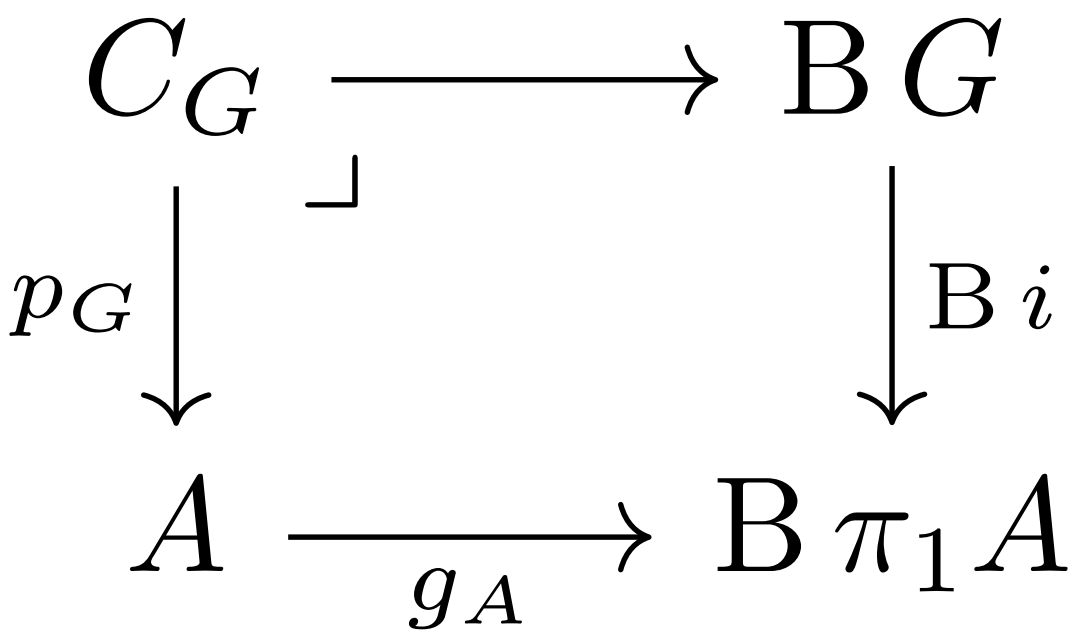
$$(f(x), |x|_1, \mathrm{refl}, p \cdot q) = (a, |x|_1, |q|_0, p)$$

with $p : \star = a$ and $q : a = f(x)$. Abstracting over a, we can suppose that q is refl by J, and we conclude immediately.

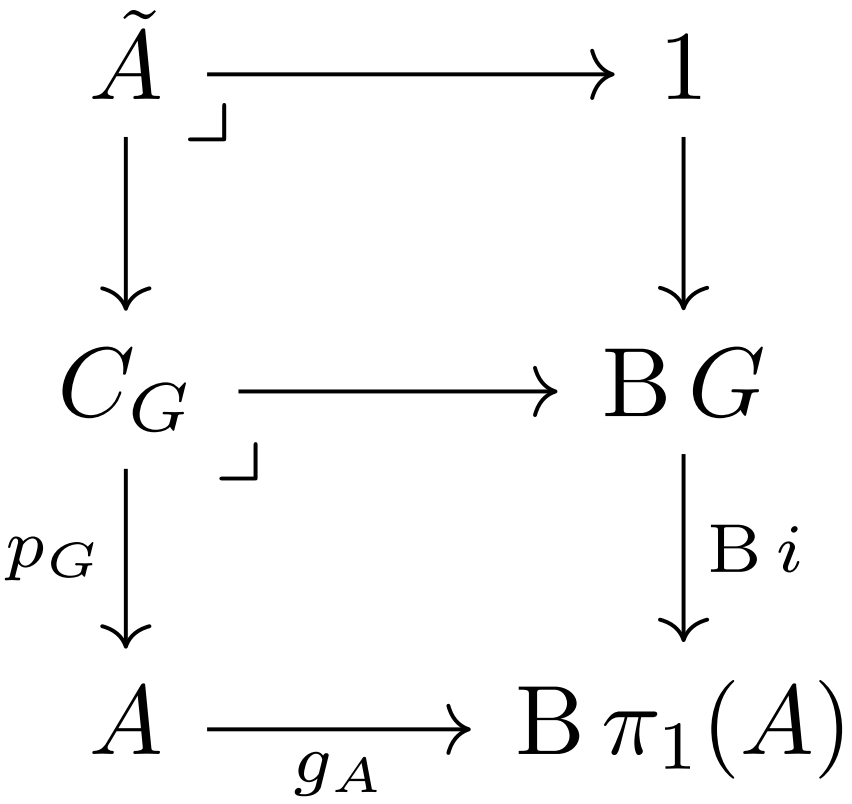
Conversely, given a subgroup $i : G \hookrightarrow \pi_1 A$, the associated subgroup (by performing the two transformations) is

$$\pi_1(p_G) : \pi_1 C_G \to \pi_1 A$$

we thus have to show that $\pi_1 C_G = G$ and the map $\pi_1 p_G = i$ (note that to be precise, we should identify the sources and the targets of the maps up to equality). We have a pullback square



which can be extended as (see above)



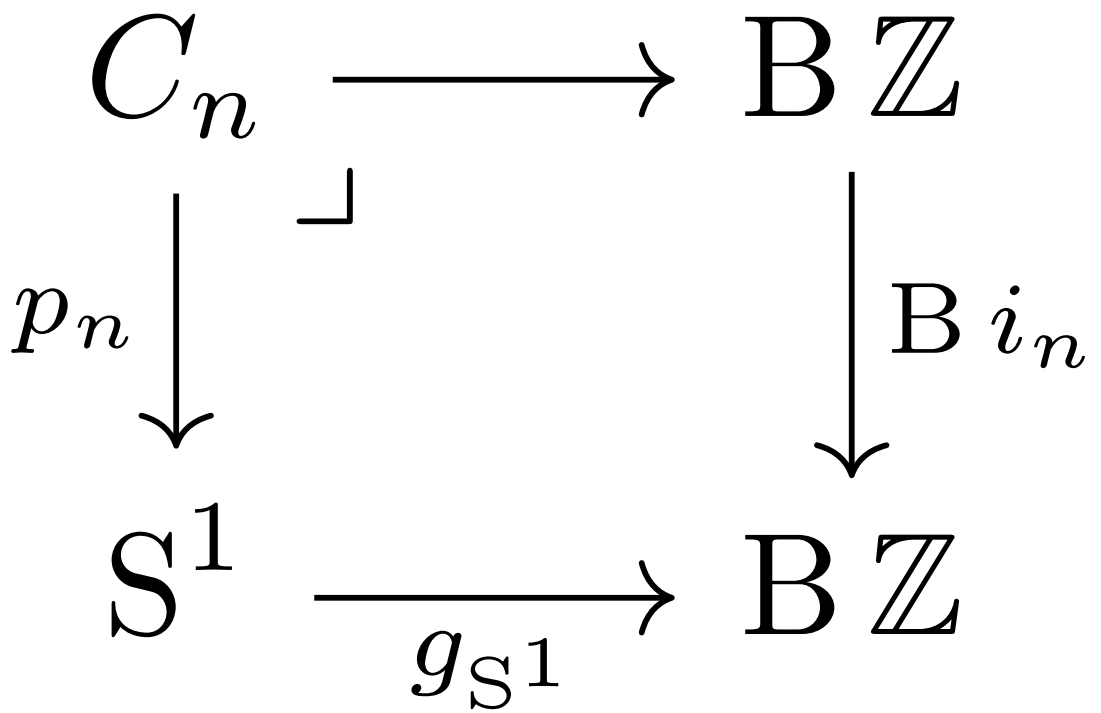
The map $C_G \to \mathrm{B}\,G$ is 1-connected because the fiber is $\tilde{A}$ which is 1-connected by Theorem 22. By [14, Lemma 7.5.14], it thus induces an equivalence $\|C_G\|_1 \simeq \|\,\mathrm{B}\,G\|_1$, and thus $\pi_1(C_G) = \pi_1(\mathrm{B}\,G) = G$. We should have the fact that $\pi_1 p_G = i$ similarly, by applying π_1 to the above square (which is not anymore a pullback but remains commutative). ◀

Example 37.

Consider the case $A \equiv \mathrm{S}^1$. The associated Galois fibration is

$$\mathrm{S}^1 \xrightarrow{|-|_1} \|\,\mathrm{S}^1\,\|_1 \equiv \mathrm{B}\,\pi_1\,\mathrm{S}^1 \simeq \mathrm{B}\,\mathbb{Z}$$

and is an equivalence, because S^1 is a groupo\operatorname{i}d. The subgroups of $\mathbb{Z}$ are of the form $i_n : \mathbb{Z} \to \mathbb{Z}$ with $i_n(k) = n \times k$ with $n > 0$ or $i_0 : 0 \to \mathbb{Z}$. And thus (connected) covering of the circle are obtained as pullbacks of the form



Since the lower arrow is an equivalence, the pullback C_n is $\mathrm{B}\,\mathbb{Z}$, i.e. S^1. And $p_n : \mathrm{S}^1 \to \mathrm{S}^1$ is the pointed map of degree n, sending the loop to loopn.

Appendix A Additional proofs

Proof of Lemma 21.

First, consider a property $P : (x, y : A) \to (p : x = y) \to \mathcal{U}$ such that $P\,x\,y\,p$ is a proposition for any $x, y : A$ and $p : x = y$. Since A is pointed connected, the following are equivalent

(i)

$P\,x\,y\,p$ for any x, y and p,

(ii)

$P \star \star p$ holds for any $p : \star = \star$.

Indeed, the second is a particular case of the first. Conversely, suppose (ii) holds, and that we are given x, y, and p. Since we want to show $P\,x\,y\,p$ which is a proposition, we can suppose given paths $p_x : \star = x$ and $p_y : \star = y$ because A is connected, we then deduce that $P\,x\,y\,p$ holds from (ii) by transport.

Now, we have that A is $(n{+}1)$-connected if and only if $x = y$ is n-connected for any $x, y : A$ (see Section 2) if and only if $\Omega\,A$ is n-connected (by the preceding observation, based on the fact that being n-connected is a proposition [14, Theorem 7.1.10]) if and only if the pointing map $a = 1 \to A$ is n-connected (because we have $\Omega\,A = \mathrm{fib}_a \star$). The reasoning is similar for the truncated version.

An alternative proof for the connected case can be found in [14, Lemma 7.5.11]. ◀

Proof of Proposition 25.

Since $\tilde{A}$ is $(n{+}1)$-connected by Theorem 22, we have $\pi_i(\tilde{A}) = 1$ for $i \leq n + 1$ [14, Lemma 8.3.2]. For $i > n + 1$, this is a consequence of the long exact sequence induced by the fiber sequence of Theorem 24, see [14, Theorem 8.4.6]:

$$\cdots \longrightarrow \pi_{i+1}\|A\|_{n+1} \longrightarrow \pi_i\tilde{A} \longrightarrow \pi_i A \longrightarrow \pi_i\|A\|_{n+1} \longrightarrow \pi_{i+1}\tilde{A} \longrightarrow \cdots$$

We have $\pi_i\|A\|_{n+1} = 1$ and the short exact sequence

$$1 \longrightarrow \pi_i \tilde{A} \longrightarrow \pi_i A \longrightarrow 1$$

shows that $\pi_i \tilde{A} = \pi_i A$. ◀

The following two lemmas (Lemma 41 and Lemma 42) are used in the subsequent proof of Lemma 35:

Lemma 41.

Suppose given an injective group morphism $f : G \to H$. Then $\ker \mathrm{B}\, f$ is a set.

Proof.

We have, by definition:

$$\ker \mathrm{B}\, f \quad \equiv \quad \Sigma(x : \mathrm{B}\, G).(\star = \mathrm{B}\, f(x))$$

Given $(x, p) : \ker \mathrm{B}\, f$, we want to prove that the loop space

$$(x, p) = (x, p)$$

is a proposition (this is enough to show that $\ker \mathrm{B}\, f$ is set by [14, Theorem 7.2.1]). Since $\mathrm{B}\, G$ is connected, we merely have a path $q : \star = x$. Since being a proposition for the loop space is a proposition, we can suppose given such a path q, and by path induction it is enough to prove that

$$(\star, p) = (\star, p)$$

is a proposition, with $p : \Omega\, \mathrm{B}\, H$. By the characterization of paths in Σ-types [14, Theorem 2.7.2] and transport along identity types [14, Theorem 2.11.3], the above type is equivalent to

$$\Sigma(a : \star = \star).(p \cdot \mathrm{B}\, f^{=}(a) = p)$$

We then have the following chain of equivalences:

$\Sigma(a : \star = \star).\quad (p \cdot \mathrm{B}\, f^{=}(a) = p)$
$= \Sigma(a : \star = \star).(\mathrm{B}\, f^{=}(a) = \mathrm{refl}_{\mathrm{B}\, f(\star)})$ by left simplification by p
$= \Sigma(a : G).(f(a) =_H 1_H)$ we use $\Omega\, \mathrm{B}\, G = G$ and $\Omega\, \mathrm{B}\, H = H$
$= \Sigma(a : G).(f(a) =_H f(1_G))$ because f is a morphism
$= \Sigma(a : G).(a =_G 1_G)$ because f is injective

This latter type is contractible [14, Lemma 3.11.8] and is thus a proposition. ◀

Lemma 42.

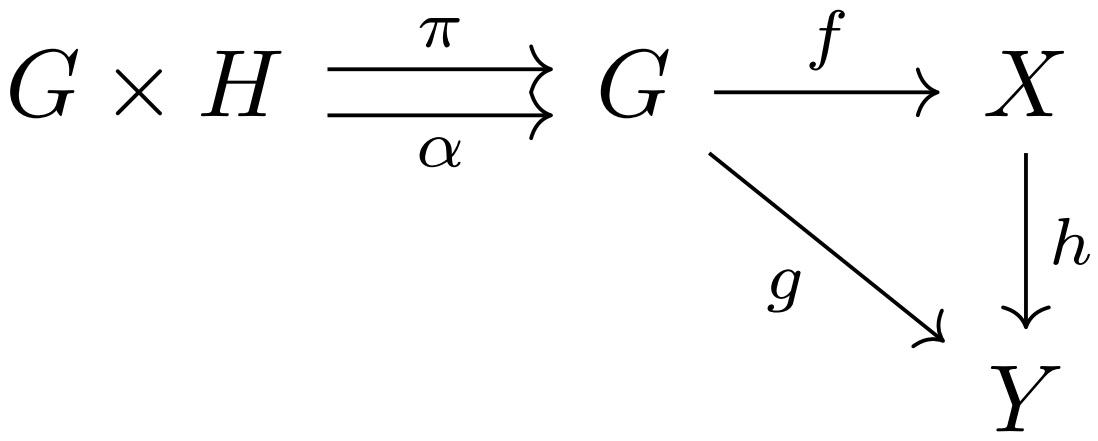
Suppose given a group morphism $i : H \to G$ and a function $f : G \to X$, where X is a set, and f is invariant by the action of H on G induced by right multiplication, and such that $\mathrm{fib}_f\, x$ is the group H equipped with its canonical action on itself for every $x : X$. Then $X = G/H$.

Proof.

Recall that the quotient is the set truncation of the homotopy coequalizer

$$G \times H \underset{\alpha}{\overset{\pi}{\rightrightarrows}} G \longrightarrow G /\!\!/ H$$

with π the first projection and α the right action of H on G (we have $\alpha(a,b) \equiv a \times i(b)$), i.e. $G/H = \|G/\!\!/H\|_0$. Otherwise said, G/H satisfies the same universal property as the above equalizer but restricted to sets. Our aim is now to show that $f : G \to X$ satisfies this universal property. The hypothesis that f is invariant under the action of H precisely means that it coequalizes the two maps. Suppose given another coequalizing map $g : G \to Y$ with Y a set. We need to show that there exists a unique map $h : X \to Y$ such that $h \circ f = g$:



The existence of such an h is implied by the fact that we have, for every $x : X$, an element $a : \mathrm{fib}_f\, x$ such that for every element $b : \mathrm{fib}_f\, x$, we have $g(a) = g(b)$. This condition is satisfied. Indeed, under the identification of $\mathrm{fib}_f\, x$ with H, we can take a to be the neutral element of H, and the fact that the action of H on itself is transitive and that g preserves this action ensures that any other image will be equal to this one. ◀

Proof of Lemma 35.

Recall that we have

$$\ker \mathrm{B}\, i \quad \equiv \quad \Sigma(x : \mathrm{B}\, H).(\star = \mathrm{B}\, i\, x)$$

We define a map

$$f : G \to \ker \mathrm{B}\, i$$

by $f(a) \equiv (\star, p_a)$ for $a : G$ corresponding to a path $p_a : \star = \star$ in $\mathrm{B}\, G$. The fibers of this map are

$\mathrm{fib}_f(x,p) \equiv \Sigma(a : G).((x,p) = (\star, p_a))$
$= \Sigma(a : G).\Sigma(q : x = \star).(p \cdot (\mathrm{B}\, i)^{=}(q) = p_a)$

In particular, for $x \equiv \star$, we have $p \equiv p_b$ for some $b : G$ and the fiber is

$\mathrm{fib}_f(\star, p_b) = \Sigma(a : G).\Sigma(c : H).(b \times i(c) = a)$
$= \Sigma(c : H).\Sigma(a : G).(b \times i(c) = a)$
$= H$

Since $\mathrm{B}\, H$ is connected, we can deduce that $\mathrm{fib}_f(x,p)$ merely is H for any $(x,p) : \ker \mathrm{B}\, i$. Furthermore, f is invariant under the action of H. Given $a : G$ and $b : H$, we have

$f(b \cdot a) \equiv (\star, p_{(b \cdot a)})$
$\equiv (\star, p_{(a \times i(b))})$
$\equiv (\star, p_a \cdot p_{i(b)})$
$= (\star, p_a)$ by transport along $p_{i(b)}^{-1}$ in $\lambda x.(\star = \mathrm{B}\, i(x))$ ([14, Th. 2.11.3])
$\equiv f(a)$

Finally, by Lemma 41, $\ker \mathrm{B}\, i$ is a set since i is injective. By Lemma 42, we deduce that $\ker \mathrm{B}\, i$ merely is G/H. ◀

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